

Technical Report on Neural Radiance Field (NeRF)

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Abstract

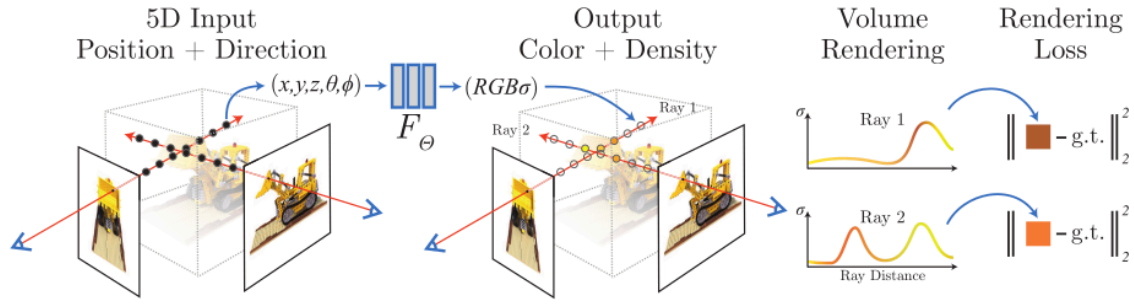
Neural Radiance Fields (NeRF) represent a novel approach to generating high-quality 3D scenes from 2D images. Leveraging neural networks, NeRF synthesises novel views of complex scenes by learning a continuous volumetric scene function. This report delves into the underlying principles of NeRF, its implementation, applications, and challenges, providing a comprehensive overview of this cutting-edge technology.

Introduction

Neural Radiance Fields (NeRF) have emerged as a significant breakthrough in computer vision and graphics, particularly in the realm of 3D scene reconstruction and novel view synthesis. Introduced by Mildenhall et al. in 2020, NeRF utilises a neural network to represent a volumetric scene, enabling the generation of photorealistic images from arbitrary viewpoints.

Principles of NeRF

NeRF encodes a 3D scene using a fully connected deep neural network that maps spatial coordinates (x, y, z) and viewing direction (θ, ϕ) to RGB colour and volume density. The core idea is to represent the scene as a continuous 5D function F_{θ} :



Volume Rendering

NeRF employs volume rendering techniques to synthesise images. The rendering equation integrates colour and density along rays cast through the scene:

$$C(\mathbf{r}) = \int_{t_n}^{t_f} T(t) \sigma(\mathbf{r}(t)) \mathbf{c}(\mathbf{r}(t), \mathbf{d}) dt, \text{ where } T(t) = \exp\left(-\int_{t_n}^t \sigma(\mathbf{r}(s)) ds\right)$$

The function $T(t)$ denotes the accumulated transmittance along the ray from t_n to t , i.e., the probability that the ray travels from t_n to t without hitting any other particle. Rendering a view from our continuous neural radiance field requires estimating this integral $C(\mathbf{r})$ for a camera ray traced through each pixel of the desired virtual camera.

We numerically estimate this continuous integral using quadrature. Deterministic quadrature, which is typically used for rendering discretized voxel grids, would effectively limit our representation's resolution because the MLP would only be queried at a fixed discrete set of locations. Instead, we use a stratified sampling approach where we partition $[t_n, t_f]$ into N evenly-spaced bins and then draw one sample uniformly at random from within each bin:

$$t_i \sim \mathcal{U} \left[t_n + \frac{i-1}{N}(t_f - t_n), t_n + \frac{i}{N}(t_f - t_n) \right].$$

Implementation

The implementation of NeRF involves several key steps:

1. **Data Collection:** Capture a set of 2D images of the scene from multiple viewpoints along with their corresponding camera parameters.
2. **Network Architecture:** Use a multi-layer perceptron (MLP) to model the 5D scene function. The network takes 3D coordinates and viewing directions as input and outputs colour and density values.
3. **Training:** Optimise the network parameters by minimising the difference between the rendered images and the actual images using a loss function, typically mean squared error (MSE).
4. **Rendering:** Once trained, the network can generate novel views by querying the scene function with new camera positions and directions.

Applications

NeRF has a wide range of applications, including:

- Virtual and Augmented Reality: Realistic 3D scene generation for immersive environments.
- Film and Animation: High-quality scene rendering for visual effects.
- Cultural Heritage: Digitization and preservation of historical sites and artefacts.

- Robotics and Autonomous Systems: Enhanced scene understanding for navigation and manipulation.

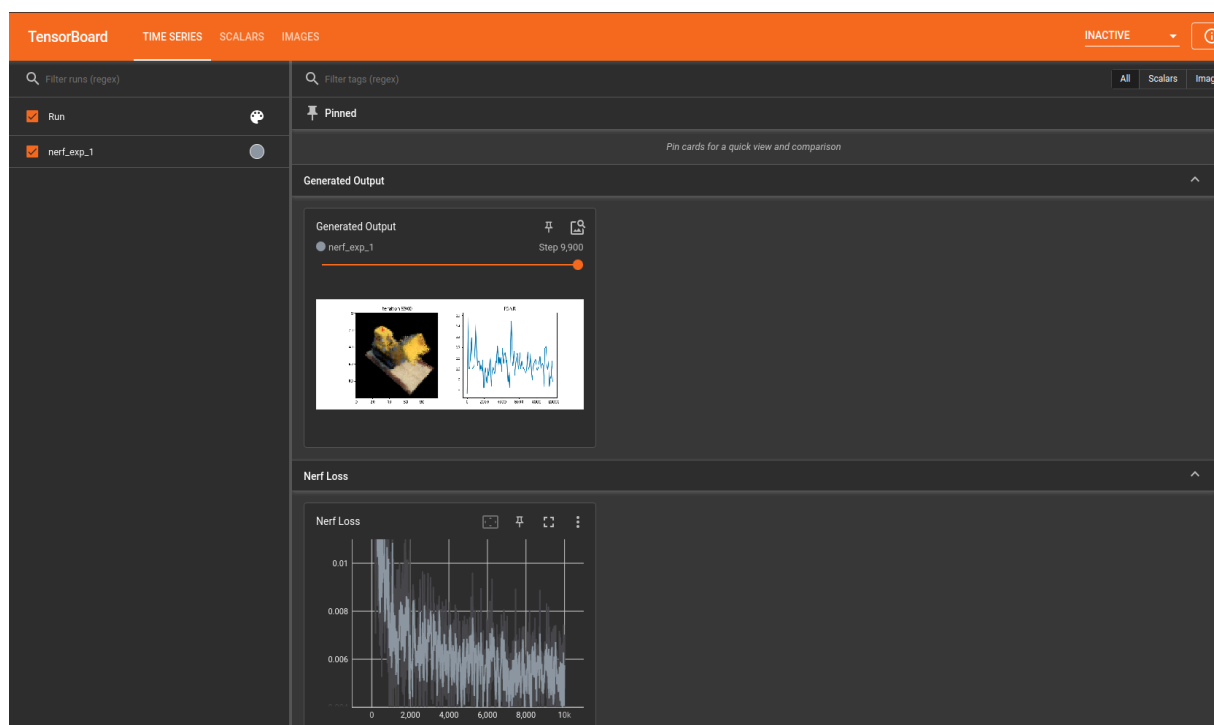
Challenges

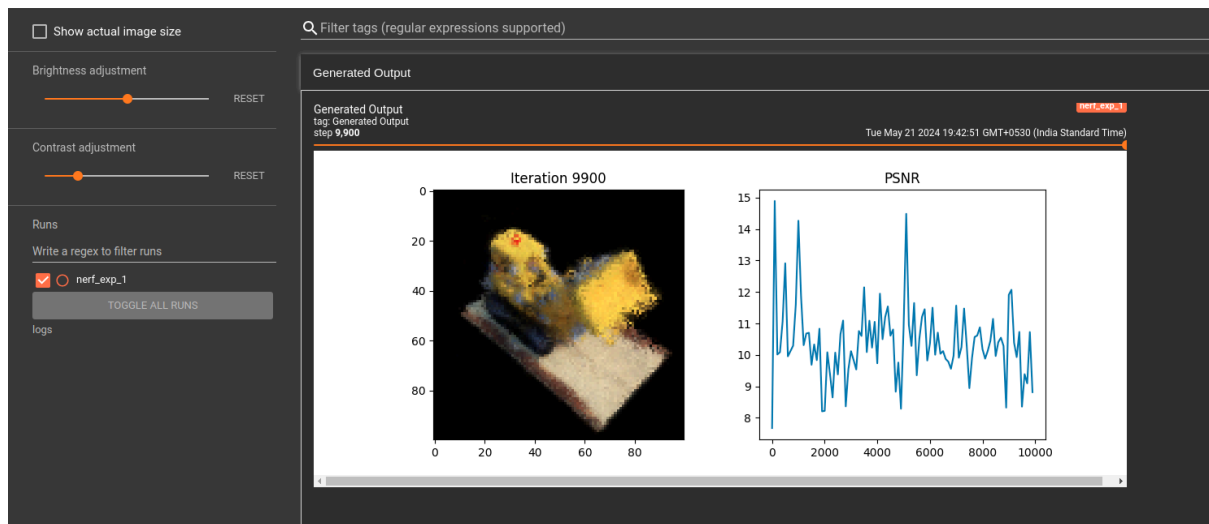
Despite its impressive capabilities, NeRF faces several challenges:

- **Computational Intensity:** Training NeRF models is computationally expensive and time-consuming.
- **Data Requirements:** High-quality results require a large number of input images with precise camera calibration.
- **Scalability:** Scaling NeRF to handle large and dynamic scenes remains a significant hurdle.
- **Generalisation:** Adapting NeRF to generalise across diverse scenes and conditions is an ongoing research area.

Reproducibility

We tried to reproduce the results of this paper using the tiny nerf model on lego dataset





Conclusion

Neural Radiance Fields represent a significant advancement in the field of 3D scene reconstruction and novel view synthesis. By leveraging the power of neural networks, NeRF provides a robust framework for generating photorealistic images from sparse 2D inputs. As research progresses, addressing current limitations will pave the way for broader adoption and new applications in various fields.

References

1. Mildenhall, B., Srinivasan, P. P., Tancik, M., Barron, J. T., Ramamoorthi, R., & Ng, R. (2020). NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis. *Proceedings of the European Conference on Computer Vision (ECCV) 2020*.